
DiMaTE: Context-Aware Malay-to-English Translation for Malaysian Regional Dialects and Slang

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Abstract

Malaysian regional dialects and slang create a practical difficulty for Malay-to-English machine translation because sentences that are natural to speakers from one state may be unfamiliar to visitors, standard-Malay speakers, or general translation systems. This project investigates whether an explicit standard-Malay bridge improves dialect-to-English translation. We construct DiMaTE-MY, a five-dialect parallel dataset covering Johor, Kedah, Kelantan, Perak, and Terengganu, with dialect Malay, standard Malay, English translation, and state labels. We compare four model levels: a direct MalayBERT baseline, a MalayBERT gold-bridge model, a two-stage MalayBERT predicted-bridge model, and an mT5 text-to-text model with direct, gold-bridge, and predicted-bridge variants. Results show that direct Malay dialect-to-English translation is the weakest setting, while bridge-based models improve translation quality. In the mT5 experiments, direct translation achieves a BERTScore F1 of 0.6695, predicted-bridge translation improves to 0.8552, and gold-bridge translation reaches 0.9334. These results support the hypothesis that standard Malay is a useful intermediate representation for translating Malaysian dialect and slang into English, while also showing that predicted-bridge systems remain limited by normalization errors.

1 Introduction

Machine translation is frequently used by travellers, international students, and non-native speakers who need to understand everyday communication. In Malaysia, these interactions do not always occur in standard Bahasa Melayu: local conversations may contain state-specific words, altered pronunciation reflected in informal spelling, and colloquial expressions. For example, the Kelantan form *kawe* can correspond to standard Malay *saya* (“I”), and dialectal word order may also differ from standard forms Khaw and Tan (2014). A general Malay-to-English translator trained mostly on standard Malay may therefore lose meaning when the input is regionally marked or informal.

Previous Malay-focused systems highlight two complementary challenges. Malay-English translation has long been affected by lexical and phrase-level ambiguity Yusoff et al. (2016). Meanwhile, Malay dialect resources remain scarce and non-standardized Khaw and Tan (2014). MalayFY demonstrates that dialect-aware assistance is practically motivated for tourists, but its rule-based dialect translation and dialect detector do not directly solve neural dialect-to-English generation Dirgantari et al. (2024).

These observations motivate a translation system that explicitly represents both the original dialect wording and its normalized standard-Malay meaning.

We study the task

$$x_d \longrightarrow y_{\text{en}},$$

where x_d is a dialect/slang Malay utterance associated with a regional label d , and y_{en} is its English translation. Our central hypothesis is that an intermediate standard-Malay representation z_{ms} provides a useful semantic bridge:

$$x_d \longrightarrow z_{\text{ms}} \longrightarrow y_{\text{en}}.$$

The bridge can reduce dialectal surface variation before English generation and makes the system interpretable: an incorrect final English translation can be traced either to dialect normalization or to Malay-to-English translation.

Our contributions are:

1. We organize a project-specific parallel dataset, DiMaTE-MY, containing dialect Malay, standard Malay, English translations, and five Malaysian state/dialect labels: Johor, Kedah, Kelantan, Perak, and Terengganu.
2. We implement a controlled model progression from direct dialect-to-English translation to gold-bridge and predicted-bridge translation, using both MalayBERT-based encoder-decoder models and a generation-native mT5 backbone.
3. We evaluate whether standard Malay functions as a useful intermediate representation by comparing direct, gold-bridge, and predicted-bridge settings under a shared train/validation/test split.
4. We show that bridge-based translation improves over direct translation, while predicted-bridge systems remain weaker than gold-bridge systems because normalization errors can propagate into the final English output.

2 Related Work

Malay dialect resources. MaDiTS was developed for Kelantanese Malay dialect translation and speech synthesis Khaw and Tan (2014). Their work emphasizes that dialectal Malay is low-resource and lacks a standardized writing convention, and it uses rules to transform standard Malay into dialectal forms. Although their goal is speech corpus construction rather than neural translation to English, the standard-Malay/dialect pairing directly motivates our bridge representation.

Malay-to-English translation. Yusoff et al. (2016) proposed a semantic-based Malay-English translation method using an n-gram model to address ambiguity beyond word-for-word translation. Their study reinforces the importance of phrase context in Malay-to-English translation. Our work addresses a related but more difficult condition: the source additionally includes regional and slang variation, and the target is produced by pretrained neural sequence generation.

Dialect-oriented applications. Dirgantari et al. (2024) present MalayFY, which provides English-to-Malay, standard-Malay-to-dialect, and dialect detection modules covering multiple Malaysian dialects. It reports a Random Forest dialect detector and a rule-driven translator. Our work differs in translation direction and design: we translate dialect Malay into English, incorporate state context in a neural model, and investigate whether standard Malay functions as an intermediate bridge.

Low-resource parallel data and multilingual generation. NusaX demonstrates how parallel data and human quality control can enable NLP evaluation for underrepresented regional languages Winata et al. (2023). Although NusaX covers Indonesian local languages rather than Malaysian dialects, it motivates our treatment of dialect-aware translation as a parallel-resource problem. Its multilingual parallel-data structure also informed the organization of our dataset, in which each dialectal Malay expression is paired with its standard-Malay normalization and English translation. In addition, its emphasis on human quality control supports our decision to prioritize carefully reviewed data and consistent train/validation/test splits. For generation, mT5 is a multilingual text-to-text Transformer pretrained on 101 languages Xue et al. (2021); it is therefore a natural stronger backbone for our generative translation setting.

Table 1: Illustration of the bridge-formatted translation task.

Field	Example
Dialect label	Kelantan
Dialect input x_d	<i>Kawe nok gi kedai.</i>
Standard Malay z_{ms}	<i>Saya hendak pergi ke kedai.</i>
English target y_{en}	<i>I want to go to the shop.</i>

B1: Direct MalayBERT baseline $[d; x_d] \rightarrow y_{en}$
M2: Gold-bridge MalayBERT $[d; x_d; z_{ms}] \rightarrow y_{en}$
M3: Predicted-bridge MalayBERT $[d; x_d] \rightarrow \hat{z}_{ms}; [d; x_d; \hat{z}_{ms}] \rightarrow y_{en}$
M4: mT5 experiments direct, gold-bridge, and predicted-bridge modes using the same inputs.

Figure 1: Implemented experimental progression. A gold bridge is an oracle condition available for analysis; a predicted bridge is deployable without manual standard-Malay input at inference time.

3 Approach

3.1 Task Formulation and Input Representation

Each example is represented as

$$(d, x_d, z_{ms}, y_{en}),$$

where d is the dialect/state label, x_d is the dialect utterance, z_{ms} is the standard-Malay normalization, and y_{en} is the reference English translation. We explicitly prepend regional context because the same surface form can have different usage across states and because state labels can help the model interpret dialect vocabulary. At inference time, the final system is expected to receive only the dialect sentence and its state label, then output the English translation; the standard-Malay field is used either as oracle bridge information or as a target for the normalizer.

3.2 Model Progression

Figure 1 summarizes the tested model family. This progression distinguishes whether improvements originate from bridge information or from a better generative backbone.

B1: Direct MalayBERT encoder-decoder baseline. MalayBERT is an encoder-only pretrained language model; the supplied baseline code constructs a generation model by placing the same MalayBERT checkpoint in the encoder and decoder of Hugging Face’s `EncoderDecoderModel`, enabling decoder cross-attention. The intended corrected source template is

$$[\text{STATE}=\{d\}] \text{ DIALECT: } \{x_d\},$$

with English as the target. This establishes whether dialect-to-English translation is possible without bridge information. It is therefore the lowest-information setting and serves as the main baseline for measuring whether bridge information actually helps.

M2: Gold standard-Malay bridge. The Level-2 model supplies the gold normalization alongside the dialect input:

$$[\text{STATE}=\{d\}] [\text{DIALECT}=\{d\}] \text{ DIALECT: } \{x_d\} \text{ STANDARD_MALAY: } \{z_{ms}\}.$$

This is an oracle experiment because the normalized Malay sentence is manually available at inference. It measures the maximum benefit of accurate bridge information rather than a fully deployable user scenario. If this model performs much better than the direct baseline, it indicates that the standard-Malay bridge contains useful information for English translation.

M3: Predicted standard-Malay bridge. The Level-3 implementation contains two MalayBERT encoder-decoder models. A normalizer predicts $\hat{z}_{ms}$ from x_d ; a translator then predicts English from x_d and $\hat{z}_{ms}$:

$$\hat{z}_{ms} = N_{\theta}(d, x_d), \quad \hat{y}_{en} = T_{\phi}(d, x_d, \hat{z}_{ms}).$$

To approximate real inference conditions, the translator should be trained with generated bridge sentences rather than always seeing the gold bridge. This prevents an unrealistic train–test discrepancy in which the translator never observes normalization errors during training. This model is more realistic than the gold-bridge setting because the standard-Malay bridge is generated automatically rather than supplied by a human annotator.

M4: Generation-native mT5 model. The Level-4 code replaces the forced MalayBERT encoder–decoder construction with `google/mt5-small`, loaded through `AutoModelForSeq2SeqLM`. mT5 naturally supports text-to-text generation and is pretrained on multilingual data Xue et al. (2021). Three modes are implemented: direct translation, gold-bridge translation, and predicted-bridge translation. The predicted-bridge mT5 variant is a deployable generation-native candidate because it combines automatic normalization with a native generative backbone. This also allows us to test whether improvements come from the bridge formulation itself, the stronger generation backbone, or both.

4 Experiments

4.1 Data

We constructed the DiMaTE-MY dataset specifically for this project to support the task of Malaysian regional dialect and slang translation into English. As existing resources for Malaysian dialect-to-English translation are limited, the dataset was designed as a parallel resource that connects dialectal Malay expressions with their normalized standard-Malay meanings and corresponding English translations. The collected workbook contains the columns *Ayat Bahasa Inggeris* (English reference), *Ayat BM* (standard-Malay normalization), *Negeri* (state/dialect label), and *Dialects Sentence* (dialect expression). We map these fields to y_{en} , z_{ms} , d , and x_d , respectively. This structure supports both direct dialect-to-English translation and the proposed bridge-based setting, in which standard Malay functions as an intermediate representation. Table 2 reports statistics computed from the supplied `dataset.xlsx` file before final cleaning.

The dataset covers five Malaysian state-associated dialect groups: Johor, Kedah, Kelantan, Perak, and Terengganu. Each state contributes 463 rows before cleaning, resulting in a balanced distribution that supports controlled comparison across dialect groups. In total, the original dataset contains 2,315 rows, including both single-word dialect vocabulary items and sentence-level travel-related expressions. Because the dataset includes short lexical items as well as longer utterances, the overall automatic evaluation scores should be interpreted together with qualitative inspection of sentence-level examples.

Before training, we remove rows with missing required fields and normalize whitespace through the preprocessing pipeline. The required fields are *ID*, *Ayat Bahasa Inggeris*, *Ayat BM*, *Negeri*, and *Dialects Sentence*. The state label is retained as a contextual feature in the model input because it may help the model distinguish region-specific vocabulary and informal expressions.

The dataset is balanced by state, which is beneficial for dialect comparison. For the final experiments, we use a shared train/validation/test split for all models so that score differences reflect model behavior rather than different data partitions. The split uses a 70/15/15 ratio, stratified by the *Negeri* column,

Table 2: Dataset audit statistics prior to cleaning.

Statistic	Value
Total rows	2,315
States/dialects covered	5
Rows per state (Johor, Kedah, Kelantan, Perak, Terengganu)	463 each
Unique English strings	447
Unique English–standard Malay pairs	456
Single-word English entries	1,050
Entries with more than three English words	990
Rows missing standard Malay	5
Rows missing dialect text	1

Table 3: Final shared split used for model comparison after removing rows with empty required fields.

Split	Rows	Ratio
Train	1,619	70%
Validation	345	15%
Test	345	15%

Table 4: Configuration in the supplied experiment code.

Setting	MalayBERT variants	mT5 variants
Epochs	3	3
Learning rate	5×10^{-5}	3×10^{-4}
Train/eval batch size	4 / 4	2 / 2
Weight decay	0.01	0.01
Warmup ratio	0.06	0.06
Maximum source length	128–192	192
Maximum target length	128	128
Beam size	4	4
Model selection metric	chrF++	chrF++

with random seed 42. Rows with empty required fields are removed before splitting, resulting in 1,619 training examples, 345 validation examples, and 345 test examples as summarized in Table 3.

4.2 Evaluation Method

We evaluate final English predictions against the gold English translations on the held-out test split shared by every system. The main comparison uses chrF++ and BERTScore F1, which were computed consistently for all completed model levels.

chrF++. chrF compares n-gram precision and recall and is particularly appropriate for morphologically and orthographically variable language generation Popović (2015). It provides a transparent surface-form score and is configured in the supplied code with word order 2.

BERTScore. BERTScore evaluates semantic similarity using contextual token embeddings rather than requiring exact token matches Zhang et al. (2020). We report BERTScore F1 using multilingual BERT, as configured in the code, to account for paraphrastic English translations.

Bridge-quality evaluation. For predicted-bridge systems, we additionally evaluate $\hat{z}_{ms}$ against the gold standard Malay using chrF++ and BERTScore. This enables error attribution: low English quality may arise because the normalizer produced an incorrect standard-Malay bridge, because the translator failed despite a good bridge, or because errors occurred in both stages.

Qualitative analysis. We inspect outputs by dialect, lexical-versus-sentence-level input, and bridge error type. Important cases include dialect pronouns, shortened colloquial forms, reordered markers, and examples where the normalized bridge is fluent but the English meaning is incorrect.

4.3 Experimental Details

All final experiments use random seed 42 and beam-search generation with four beams. Table 4 summarizes the main training settings used across the implemented systems. The MalayBERT variants use `mesolitica/bert-base-standard-bahasa-cased`; the mT5 experiments use `google/mt5-small`.

Reproducibility setup. For the final experiments, all models use the same corrected task direction: `Dialects Sentence` is the source input and `Ayat Bahasa Inggeris` is the English target. The shared split is generated once using a 70/15/15 train/validation/test ratio, stratified by `Negeri`, after dropping rows with empty required fields. This ensures that direct, gold-bridge, and predicted-bridge

Table 5: Final English translation results on the shared held-out test set. The best overall oracle result is highlighted in bold.

Model	chrF++	BERTScore F1
B1 MalayBERT direct	11.19	0.6392
M2 MalayBERT + gold bridge	56.79	0.8932
M3 MalayBERT + predicted bridge	57.88	0.8594
M4 mT5 direct	10.44	0.6695
M4 mT5 + gold bridge	64.62	0.9334
M4 mT5 + predicted bridge	38.25	0.8552

Table 6: Normalization quality for predicted-bridge models.

Normalizer	chrF++	BERTScore F1
MalayBERT normalizer (M3)	67.00	0.8733
mT5 normalizer (M4 predicted bridge)	49.29	0.8731

systems are compared on the same held-out test examples. Model checkpoints are selected using validation chrF++, while final reported scores are computed separately on the held-out test split.

4.4 Results

Table 5 reports the final English translation results on the shared held-out test set. Both direct translation models obtain the lowest scores, while all bridge-based variants achieve higher performance. The best overall result is obtained by **M4 mT5 + gold bridge**, which reaches a chrF++ score of 64.62 and a BERTScore F1 of 0.9334. Since this model uses the reference standard-Malay bridge, it is reported as an oracle setting. Among the predicted-bridge models, which generate the intermediate standard-Malay sentence automatically, **M3 MalayBERT + predicted bridge** achieves the highest scores, with chrF++ 57.88 and BERTScore F1 0.8594.

Table 6 reports the normalization quality of the two predicted-bridge models. The MalayBERT normalizer obtains a higher chrF++ score of 67.00, compared with 49.29 for the mT5 normalizer. Their BERTScore F1 scores are nearly identical, at 0.8733 and 0.8731, respectively. These results evaluate the intermediate standard-Malay predictions before they are used for English translation.

5 Analysis

5.1 Effect of the Standard-Malay Bridge

The results support the standard-Malay bridge hypothesis. Both direct models perform weakest: the MalayBERT direct baseline obtains a BERTScore F1 of 0.6392, while the mT5 direct model obtains 0.6695. These low scores suggest that direct dialect-to-English translation is difficult because the model must handle dialectal spelling, slang, regional vocabulary, and English generation in a single step.

Gold-bridge models perform substantially better because they receive the reference standard-Malay sentence as additional semantic information. M2 reaches a BERTScore F1 of 0.8932, while M4 gold bridge reaches 0.9334. This shows that when the dialect meaning is first expressed in standard Malay, the English translation task becomes much easier. The gold-bridge setting is therefore useful as an upper-bound experiment, although it is not fully deployable because it assumes access to a human-written standard-Malay bridge at inference time.

Predicted-bridge models are more realistic because they generate the standard-Malay bridge automatically. M3 obtains a translation BERTScore F1 of 0.8594, and M4 predicted bridge obtains 0.8552. Both are clearly better than their direct counterparts, which shows that automatic normalization helps. However, both remain below the gold-bridge models, indicating that errors in the predicted standard-Malay bridge can propagate into the final English translation.

Table 7: Relative improvement from direct translation to bridge-based translation using BERTScore F1.

Comparison	BERTScore F1 change	Interpretation
B1 direct to M2 gold bridge	+0.2540	Gold bridge strongly improves MalayBERT translation
B1 direct to M3 predicted bridge	+0.2202	Predicted bridge improves MalayBERT translation
M4 direct to M4 gold bridge	+0.2639	Gold bridge strongly improves mT5 translation
M4 direct to M4 predicted bridge	+0.1857	Predicted bridge improves mT5 translation

5.2 MalayBERT versus mT5

The MalayBERT and mT5 models show different strengths. For direct translation, both models perform poorly, with mT5 slightly higher in BERTScore F1 but similarly low in chrF++. This suggests that simply changing the backbone is not enough to solve dialect-to-English translation without bridge information.

In the gold-bridge setting, mT5 performs best overall, with a BERTScore F1 of 0.9334 compared with 0.8932 for the MalayBERT gold-bridge model. This is expected because mT5 is a generation-native text-to-text model, while MalayBERT is originally an encoder-only model adapted into an encoder-decoder structure. When clear bridge information is available, mT5 appears better able to convert the enriched input into fluent English.

For predicted-bridge translation, M3 and M4 obtain very similar BERTScore F1 scores: 0.8594 and 0.8552. This suggests that both systems preserve similar semantic meaning in their final English outputs. However, M3 obtains higher chrF++, which indicates closer surface-level overlap with the reference translations. One possible explanation is that the M3 normalizer also has higher normalization chrF++, producing standard-Malay bridges that are closer in wording to the reference bridge.

5.3 Normalization and Error Propagation

The normalization results help explain the predicted-bridge behavior. The M3 normalizer achieves a chrF++ score of 67.00 and BERTScore F1 of 0.8733, while the M4 normalizer achieves chrF++ 49.29 and BERTScore F1 0.8731. The nearly identical BERTScore F1 suggests that both normalizers often preserve the intended meaning, but the higher chrF++ for M3 suggests stronger surface-form similarity to the reference standard Malay.

This matters because the predicted bridge becomes part of the translator input. If the normalizer produces a bridge that is semantically correct but phrased differently from the training references, BERTScore may remain high while chrF++ decreases. If the normalizer produces an incorrect bridge, the translator may generate an incorrect English sentence even if the translator itself is capable. This explains why predicted-bridge models improve over direct translation but do not reach gold-bridge performance.

5.4 Limitations

Several limitations remain. First, the dataset is relatively small for neural translation and contains many single-word entries, which may not fully represent natural conversational translation. Second, the split is stratified by state but still may contain repeated or semantically similar English-Malay pairs across splits. This means final scores should be interpreted as controlled project results rather than broad real-world generalization. Third, the gold-bridge setting is an oracle experiment: it is useful for measuring the value of standard-Malay information, but it is not directly deployable without a normalizer. Finally, automatic metrics such as chrF++ and BERTScore cannot fully judge dialect meaning preservation, so future work should include human evaluation by speakers familiar with each dialect.

6 Conclusion

This project investigates context-aware Malay dialect-to-English translation using a standard-Malay bridge. We construct a balanced five-state dataset covering Johor, Kedah, Kelantan, Perak, and Terengganu, and compare direct translation, gold-bridge translation, and predicted-bridge translation across MalayBERT-based and mT5-based systems. The final results show that direct dialect-to-English translation is the weakest setting, while bridge-based translation substantially improves performance.

The strongest overall result is obtained by the mT5 gold-bridge model, which reaches a BERTScore F1 of 0.9334. This confirms that standard Malay provides useful semantic information for English translation when the bridge is accurate. The predicted-bridge systems are more realistic because they generate the standard-Malay bridge automatically, and both MalayBERT and mT5 predicted-bridge models improve clearly over their direct counterparts. However, they remain below the gold-bridge upper bound, showing that normalization errors can propagate into the final English output.

Overall, the findings support the main hypothesis that standard Malay is a useful intermediate representation for Malaysian dialect and slang translation. The results also show that future improvement should focus on better dialect-to-standard-Malay normalization, stronger handling of sentence-level examples, and human evaluation by speakers familiar with each dialect.

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Dialect	English Reference	Standard Malay	Dialect Sentence
Johor	Where can I buy tickets?	Di mana saya boleh membeli tiket?	Kat mana aku boleh beli tiket?
Kedah	Yes, this is my first time here.	Ya, ini kali pertama saya di sini.	Ha'ah, ni kali pertama saya mai sini.
Kelantan	Please let me know when we arrive at the next station.	Tolong beritahu saya apabila kita sudah sampai ke stesen seterusnya.	Tolong royak ko ambo bilo sampai stesen seterusnya.
Perak	Yes, I am allergic to peanuts.	Ya, saya alah kepada kacang tanah.	Ye, teman alah kacang tanah.
Terengganu	I do not think that I can eat this durian.	Saya tak rasa saya boleh makan durian ini.	Saya rase saya tak boleh makang duriang ni.

Table 8: Representative parallel examples from the DiMaTE-MY dataset across the five Malaysian regional dialect categories.

A Additional Analysis

A.1 Representative Dataset Examples

Table 8 presents representative parallel examples from the DiMaTE-MY dataset. Each entry contains an English reference sentence, its standard-Malay equivalent, and a corresponding regional dialect expression. These examples illustrate the lexical and surface-form variation that motivates the use of an intermediate standard-Malay bridge in our translation framework.

The examples in Table 8 demonstrate that regional dialect expressions may differ from standard Malay in pronouns, discourse markers, vocabulary, and spelling conventions. For example, the Perak sentence uses *teman* in place of standard Malay *saya*, while the Kelantan example uses dialect forms such as *royak*, *ambo*, and *bilo*. Similarly, the Terengganu example includes the surface forms *rase*, *makang*, and *duriang*, which differ orthographically from their standard-Malay equivalents.

These variations support the motivation for using standard Malay as an intermediate representation. Instead of requiring a translation model to map every regional surface form directly into English, the proposed bridge-based framework first attempts to recover a normalized standard-Malay meaning before generating the final English translation.

During dataset inspection, some rows were found to require additional semantic-alignment checking because the dialect sentence did not always express the same meaning as the corresponding English and standard-Malay references. Therefore, meaning consistency should be verified during preprocessing, and misaligned entries should be removed before final model training and evaluation.

A.2 Error Analysis Categories

To better understand the remaining weaknesses of the deployable predicted-bridge system, we categorize representative translation errors according to their likely source. Table 9 summarizes the main error types considered in our analysis.

The error categories in Table 9 are particularly relevant to dialect translation because non-standard forms are not always recoverable through surface matching alone. In our setting, a model may produce fluent English while still failing to preserve the original dialect meaning. For this reason, future work should supplement automatic metrics with human evaluation by speakers familiar with the relevant Malaysian dialects.

Error Type	Description	Expected Effect on Translation
Dialect pronoun error	The model fails to normalize state-specific pronouns such as <i>kawe</i> .	The English output may assign the wrong speaker or produce an incomplete sentence.
Slang vocabulary error	A colloquial or shortened regional word is mistranslated or omitted.	The main intent of the utterance may be weakened or changed.
Bridge omission	Important content in the dialect input is absent from the predicted standard-Malay bridge.	The final English translation may be fluent but semantically incomplete.
Surface-form variation	The predicted bridge expresses the correct meaning using wording different from the reference bridge.	BERTScore may remain high while chrF++ decreases because of reduced exact overlap.
Error propagation	An incorrect normalization is passed directly into the English translator.	The English output inherits the semantic error produced during normalization.

Table 9: Error categories used for qualitative analysis of predicted-bridge translation outputs.

B Reproducibility and Team Contributions

B.1 Reproducibility Details

The experiments in this study compare direct dialect-to-English translation, gold-bridge translation, and predicted-bridge translation. The overall predicted-bridge training and evaluation procedure is summarized below:

1. The dataset is cleaned by removing samples with empty required fields and normalizing whitespace.
2. Each sample is represented using a dialect sentence, a standard-Malay normalization, an English target sentence, and a state/dialect label.
3. A dialect-to-standard-Malay normalizer is trained using the training portion of the dataset.
4. The trained normalizer generates predicted standard-Malay bridge sentences.
5. A translation model is trained to generate English from the dialect input together with its bridge representation.
6. Model checkpoints are selected using validation chrF++.
7. Final English translation performance is evaluated on the held-out test set using chrF++ and BERTScore F1.

A stricter future evaluation may assign all dialect realizations associated with the same English–standard-Malay meaning pair to the same data split. Such a grouped splitting procedure would prevent semantically equivalent targets from appearing in both the training and test portions of the dataset through different regional dialect variants.

B.2 Team Member Contributions

Table 10 summarizes the contributions of each team member. All members participated in discussion of the research problem, interpretation of the results, and preparation of the final project submission.

Team Member	Contribution
Ahmad Fahreen Qusyairi bin Md Yusof	Designed the MalayBERT baseline and the M3 predicted standard-Malay bridge model; performed data preprocessing; and trained the M3 predicted-bridge model.
Arif Hakimi Bin Mohd Nurulfaizin	Designed the M2 gold standard-Malay bridge model and the M4 generation-native mT5 model; performed data preprocessing; and trained the M4 generation-native mT5 model.
Nur Asha Syafiqah Binti Suhaimi	Contributed to data generation and data pre-/post-processing; and trained the MalayBERT baseline model.
Ahmad Naufal Bin Ahmad Asro	Contributed to data generation and data pre-/post-processing; and trained the M2 gold standard-Malay bridge model.

Table 10: Summary of team member contributions to the DiMaTE project.